

ST5209/X Assignment 4

Teng Siyuan

Set up

1. Make sure you have the following installed on your system: $\text{\LaTeX}$, R4.2.2+, RStudio 2023.12+, and Quarto 1.3.450+.
2. Pull changes from the course [repo](#).

Submission

1. Render the document to get a .pdf printout.
2. Submit both the .qmd and .pdf files to Canvas.

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

1. AR polynomial

Consider the AR(2) model

$$X_t = 4 + 0.5X_{t-1} - 0.25X_{t-2} + W_t. \quad (1)$$

- a. What is the autoregressive polynomial?

$$\phi(z) = 1 - 0.5z + 0.25z^2$$

b. What are its roots?

Let

$$\phi(z) = 0$$

We have

$$z = 1 \pm \sqrt{3}i$$

c. Is this model causal? Why?

The model is causal since all roots are outside the unit circle.

d. Given the representation

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j W_{t-j},$$

solve for $\psi_0, \psi_1, \psi_2, \psi_3$. *Hint: Follow the strategy discussed in `linear_process_coefs.qmd`.*

Using $\phi(z) \left(\sum_{j=0}^{\infty} \psi_j z^j \right) = 1$

We have

$$\begin{aligned} (1 - 0.5z + 0.25z^2)(\psi_0 + \psi_1 z + \psi_2 z^2 + \psi_3 z^3 + \dots) \\ = \psi_0 + (\psi_1 - 0.5\psi_0)z + (\psi_2 - 0.5\psi_1 + 0.25\psi_0)z^2 + (\psi_3 - 0.5\psi_2 + 0.25\psi_1)z^3 + \dots = 1 \end{aligned}$$

Then we have

$$\begin{cases} \psi_0 = 1 \\ \psi_1 - 0.5\psi_0 = 0 \\ \psi_2 - 0.5\psi_1 + 0.25\psi_0 = 0 \\ \psi_3 - 0.5\psi_2 + 0.25\psi_1 = 0 \end{cases}$$

which gives

$$\begin{cases} \psi_0 = 1 \\ \psi_1 = 0.5 \\ \psi_2 = 0 \\ \psi_3 = -0.125 \end{cases}$$

2. Likelihood

Consider the AR(1) model

$$X_t = 1 - 0.6X_{t-1} + W_t,$$

where $W_t \sim WN(0, 0.25)$ is Gaussian white noise. We are given observations $x_1 = 0.2, x_2 = -0.3, x_3 = 0.4$.

- a. What is the mean of X_t ?

We have

$$\mu = 1 - 0.6\mu$$

which gives

$$\mu = 0.625$$

- b. Write the conditional likelihood of this model, when conditioning on the value of x_1 .

We have

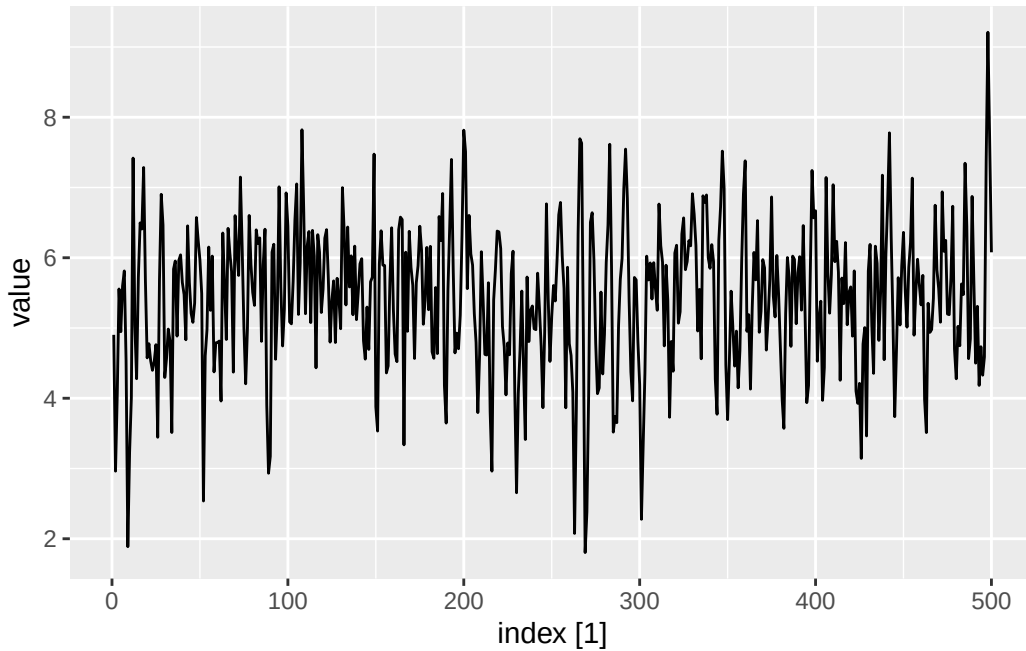
$$X_t | X_{t-1} = x_{t-1} \sim N(1 - 0.6x_{t-1}, 0.25)$$

$$\begin{aligned} L(\theta | x_1) &= \prod_{t=2}^3 \frac{1}{\sqrt{2\pi \cdot 0.25}} \exp\left(-\frac{(x_t - (1 - 0.6x_{t-1}))^2}{2 \times 0.25}\right) \\ &= \frac{1}{2\pi \times 0.25} \exp\left(-\frac{(-0.3 - 0.88)^2 + (0.4 - 1.18)^2}{0.5}\right) = 0.01164 \end{aligned}$$

3. Reversibility

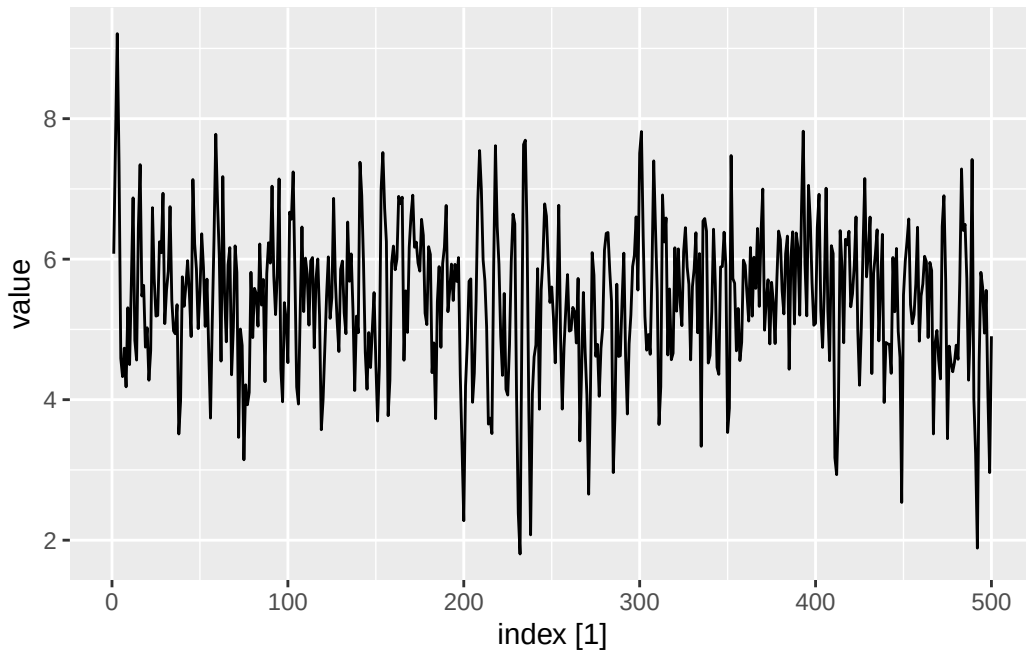
- a. Using `set.seed(5209)` to fix a random seed, create a sample trajectory of length $n = 500$ from the AR(2) model from Problem 1 using `arma.sim()`.

```
set.seed(5209)
ar2<-arma.sim(model=list(ar=c(0.5,-0.25)),n=500,mean=4)|>
  as_tsibble()
ar2|> autoplot(.vars = value)
```



- b. Reverse the time index of the vector you obtain using `rev()`.

```
ar2_rev<-rev(ar2$value)|>
  ts()|>
  as_tsibble()
ar2_rev|>autoplot(.vars=value)
```



- c. Fit an AR(2) model to the reversed time series using `fable`. Hint: You may use the code snippet `model(AR(X ~ order(2)))`.

```
fit<-ar2_rev|>
  model(AR(value~order(2)))
```

- d. Inspect the model parameters using `tidy()`. Why are they similar to the those in Equation 1?

```
tidy(fit)
```

A tibble: 3 x 6

	.model	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	AR(value ~ order(2))	constant	4.01	0.260	15.4	5.35e-44
2	AR(value ~ order(2))	ar1	0.499	0.0433	11.5	2.47e-27
3	AR(value ~ order(2))	ar2	-0.239	0.0436	-5.49	6.54e- 8

The parameters are similar to those in Equation 1 because of the reversibility of stationary AR process.

- e. Make a forecast with $h = 10$. What does this correspond to in terms of the original time series?

```
fit|>
  forecast(h=10)
```

A fable: 10 x 4 [1]
Key: .model [1]

	.model	index
	<chr>	<dbl>
1	AR(value ~ order(2))	501
2	AR(value ~ order(2))	502
3	AR(value ~ order(2))	503
4	AR(value ~ order(2))	504
5	AR(value ~ order(2))	505
6	AR(value ~ order(2))	506
7	AR(value ~ order(2))	507
8	AR(value ~ order(2))	508
9	AR(value ~ order(2))	509
10	AR(value ~ order(2))	510

i 2 more variables: value <dist>, .mean <dbl>

This correspond to predicting the 10 values that came before x_1 in the original series.

4. Yule-Walker

- a. Write the Yule-Walker equations for the AR(2) model from Problem 1.

$$\begin{cases} k = 0 : & \gamma(0) = 0.5\gamma(1) - 0.25\gamma(2) + \sigma^2 \\ k = 1 : & \gamma(1) = 0.5\gamma(0) - 0.25\gamma(1) \\ k = 2 : & \gamma(2) = 0.5\gamma(1) - 0.25\gamma(0) \end{cases}$$

- b. Arrange the equations in the following matrix form (i.e. fill in the missing entries):

$$\begin{bmatrix} 1 & -0.5 & 0.25 \\ -0.5 & 1.25 & 0 \\ 0.25 & -0.5 & 1 \end{bmatrix} \begin{bmatrix} \gamma(0) \\ \gamma(1) \\ \gamma(2) \end{bmatrix} = \begin{bmatrix} \sigma^2 \\ 0 \\ 0 \end{bmatrix}.$$

- c. Solve the system from part b) for $\gamma(0), \gamma(1), \gamma(2)$ numerically using `solve()`.

```
A <- matrix(c(
  1,    -0.5,   0.25,
 -0.5,   1.25,   0,
  0.25, -0.5,   1
), nrow = 3, byrow = TRUE)

b <- c(1, 0, 0) # sigma^2 = 1

solve(A, b)
```

```
[1] 1.26984127 0.50793651 -0.06349206
```

- d. Given

$$\Gamma_2 = \begin{bmatrix} \gamma(0) & \gamma(1) \\ \gamma(1) & \gamma(0) \end{bmatrix},$$

what is the top left entry of Γ_2^{-1} in terms of $\gamma(0)$ and $\gamma(1)$?

We have

$$\Gamma_2^{-1} = \frac{1}{\gamma(0)^2 - \gamma(1)^2} \begin{bmatrix} \gamma(0) & -\gamma(1) \\ -\gamma(1) & \gamma(0) \end{bmatrix}$$

The top left entry is

$$\frac{\gamma(0)}{\gamma(0)^2 - \gamma(1)^2}$$

- e. Write a 95% confidence interval for ϕ_1 using your answers for d) and 3d).

```
phi_0<-tidy(fit)$estimate[2]
gamma0 <- solve(A,b)[1]
gamma1 <- solve(A,b)[2]
sigma2<-1
sd<- sqrt((sigma2*gamma0/(gamma0**2-gamma1**2))/500)
q<-qnorm(0.975)
c(phi_0-q*sd,phi_0+q*sd)
```

```
[1] 0.4138685 0.5836063
```

5. Recursive forecasting for ARMA(1, 1)

Consider the $ARMA(1,1)$ model

$$X_t - 0.5X_{t-1} = W_t + 0.5W_{t-1}.$$

In this question, we will investigate recursive forecasting. The following code snippet generates a sequence of length $n = 50$ drawn from the above model.

```
set.seed(5209)
n <- 50
wn <- rnorm(n)
xt <- arima.sim(model = list(ar = 0.5, ma = 0.5), innov = wn, n = n)
```

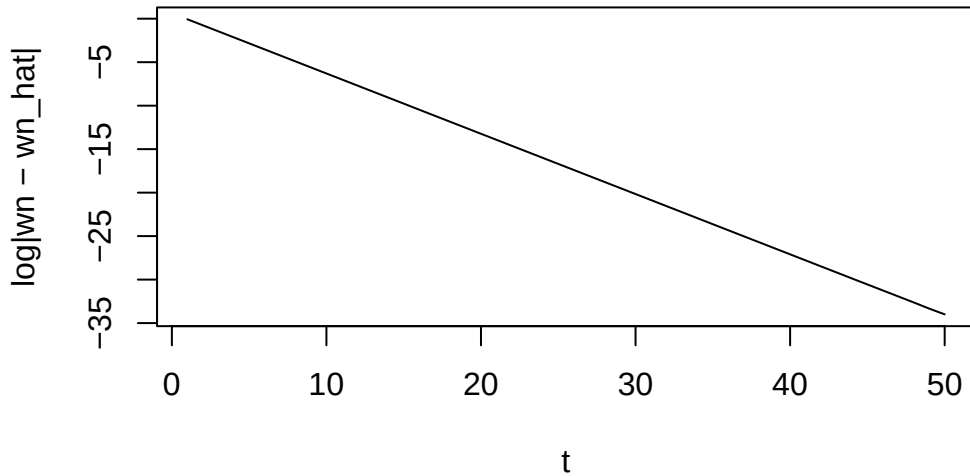
- a. Fill in the following code snippet using equation (12.14) to generate a sequence `wn_hat`.

```
wn_hat <- rep(0, n)
wn_hat[[1]] <- xt[[1]]
for (i in 2:n) {
  wn_hat[i]=xt[i]-0.5*xt[i-1]-0.5*wn_hat[i-1]
}
```

- b. Make a time plot of the log absolute difference between `wn` and `wn_hat`.

```
log_diff <- log(abs(wn - wn_hat))
plot(1:n, log_diff, type = "l",
     xlab = "t", ylab = "log|wn - wn_hat|",
     main = "Log Absolute Difference between wn and wn_hat")
```

Log Absolute Difference between wn and wn_hat



- c. What consequence does this have for truncated forecasts?

The forecasts become more and more precise as t increase, suggesting the initialization error decays over time.

- d. Compute the truncated forecast for X_{53} .

```
X_tilde_51 <- 0.5 * xt[50] + 0.5 * wn_hat[50]
X_tilde_52 <- 0.5 * X_tilde_51
X_tilde_53 <- 0.5 * X_tilde_52

cat("Truncated forecast for X_53 =", X_tilde_53, "\n")
```

Truncated forecast for $X_{53} = 0.1584008$

6. ACF, PACF, and BLPs

Let (X_t) be a mean zero stationary process with the following autocovariance values:

$$\gamma_X(0) = 2, \gamma_X(1) = 1.4, \gamma_X(2) = 0.6, \gamma_X(3) = 0.4, \gamma_X(4) = 0.2.$$

- a. Can (X_t) be an MA(2) process? Explain why or why not.

No. For MA(2) process, there should be cuts off to zero for ACVF after lag q . Since $\gamma_X(3) = 0.4 \neq 0$, it can't be MA(2) process.

- b. Can (X_t) be an AR(1) process? Explain why or why not.

No. For AR(1) process, the ACVF satisfies $\gamma(k) = \phi^k \gamma(0)$, however, we can find that $\frac{\gamma_X(4)}{\gamma_X(3)} \neq \frac{\gamma_X(3)}{\gamma_X(2)}$, which indicates this can't be AR(1) process.

- c. What is the best linear predictor $\hat{X}_4$ for X_4 given only $X_3 = 2$?

$$\hat{X}_4 = \frac{\gamma_X(1)}{\gamma_X(0)} X_3 = 1.4$$

- d. Using the notation in part c), what is the variance of $X_4 - \hat{X}_4$?

$$\text{Var}(X_4 - \hat{X}_4) = \text{Var}(X_4 - \frac{\gamma_X(1)}{\gamma_X(0)} X_3) = [1 + (\frac{\gamma_X(1)}{\gamma_X(0)})^2] \gamma_X(0) - 2 \frac{\gamma_X(1)}{\gamma_X(0)} \gamma_X(1) = 1.02$$

- e. What is the best linear predictor $\hat{X}_4$ for X_4 given only $X_2 = 2$?

$$\hat{X}_4 = \frac{\gamma_X(2)}{\gamma_X(0)} X_2 = 0.6$$

- f. Using the notation in part e), what is the variance of $X_4 - \hat{X}_4$?

$$\text{Var}(X_4 - \hat{X}_4) = \text{Var}(X_4 - \frac{\gamma_X(2)}{\gamma_X(0)} X_2) = [1 + (\frac{\gamma_X(2)}{\gamma_X(0)})^2] \gamma_X(0) - 2 \frac{\gamma_X(2)}{\gamma_X(0)} \gamma_X(2) = 1.82$$

- g. Let α_X denote the partial autocorrelation function of (X_t) . What is $\alpha_X(1)$?

$$\alpha_X(1) = \frac{\gamma_X(1)}{\gamma_X(0)} = 0.7$$

- h. What is $\alpha_X(3)$?

We use the Yule-Walker equations because PACF at lag h is by definition the last coefficient ϕ_h of the best linear predictor of X_t given $X_{t-1}, \dots, X_{t-h}$, and the optimality condition of this predictor directly yields the Yule-Walker system $\Gamma_h \phi = \gamma_{1:h}$

$$\begin{bmatrix} \gamma(0) & \gamma(1) & \gamma(2) \\ \gamma(1) & \gamma(0) & \gamma(1) \\ \gamma(2) & \gamma(1) & \gamma(0) \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \end{bmatrix} = \begin{bmatrix} \gamma(1) \\ \gamma(2) \\ \gamma(3) \end{bmatrix}$$

The PACF at lag 3 is defined as the third entry of $\Gamma_3^{-1}\gamma_{1:3}$

$$\Gamma_3 = \begin{bmatrix} 2 & 1.4 & 0.6 \\ 1.4 & 2 & 1.4 \\ 0.6 & 1.4 & 2 \end{bmatrix}, \quad \gamma_{1:3} = \begin{bmatrix} 1.4 \\ 0.6 \\ 0.4 \end{bmatrix}$$

```
Gamma3 <- matrix(c(2, 1.4, 0.6, 1.4, 2, 1.4, 0.6, 1.4, 2), nrow = 3, byrow = TRUE)
gamma13 <- c(1.4, 0.6, 0.4)
phi <- solve(Gamma3, gamma13)
phi[3]
```

```
[1] 0.3928571
```